EEG Paradigms for an 8-Channel BCI

Feasibility, Electrode Selection, Training Requirements, and the Path to Task Control

biopipe / Hybrid EEG-EMG BMI • Design report for the EEG training module • Hardware: OpenBCI Cyton (8 channels, ADS1299, 250 Hz) + Ultracortex • 19 September 2026

Purpose. This report assesses which EEG paradigms are worth building into biopipe's EEG training module given a strictly 8-channel montage, and recommends where to start. For each paradigm it covers what is decoded, feasibility at 8 channels, the best electrode placement, the data and time needed to train a usable model, and reputable references. The eventual goal is discrete **task classification** to control an external system, so paradigms are ultimately judged on control value, not decoding novelty.

1. Scope and constraints

• **8 data channels** on one Cyton (ADS1299, 24-bit, 250 Hz over the RF dongle), plus SRB reference and BIAS. No room for a dense montage, so electrodes must be spent where each paradigm's signal is strongest.

• **Dry/comb Ultracortex electrodes** raise noise and impedance versus wet gel, which lowers the effective SNR ceiling — this penalises low-amplitude cognitive signals (emotion, single-trial P300) more than high-contrast ones (SSVEP, occipital alpha).

• **Target = discrete commands** (e.g. select / left / right / stop) fused with the working EMG decoder. Continuous or high-dimensional decoding is out of scope.

• **Practical training budget**: a few subjects and short calibration sessions, classical models that train in seconds on a CPU, with deep models only where pre-training data exists.

2. Paradigm summary

Feasibility and control-fit are rated Low / Medium / High for this specific 8-channel, dry-electrode rig.

Paradigm	Best channels (max 8)	8-ch feasibility	Training burden	Control fit
Motor imagery (MI)	C3, Cz, C4 (+FC/CP)	Medium-High	Per-subject calibration; 2-class strong, 4-class harder	High
ERD/ERS	C3, Cz, C4	High (2-class)	Small; it is the mechanism behind MI	High (gating / go)
Emotion (valence/arousal)	Fp1, Fp2, F3, F4 (+T/P)	Low-Medium	Large; elicitation protocol; subject-specific	Low
SSVEP	O1, O2, Oz (+POz)	High	Near zero (CCA is calibration-free)	High (menu select)
P300	Pz, Cz, Fz (+P3/P4)	Medium	Short; needs stimulus repetitions	Medium (speller/menu)
Occipital alpha	O1, O2	High	Trivial (threshold)	Medium (idle / switch)
Error potentials (ErrP)	FCz, Cz	Medium	Moderate; corrective signal	High (as a safety layer)

3. Motor imagery (MI)

What it decodes. Imagined movement of the hands/feet suppresses the sensorimotor mu (8-13 Hz) and beta (13-30 Hz) rhythms contralaterally (see ERD/ERS, §4). Classes are typically left hand / right hand / feet / rest.

Best channels. C3 and C4 (hand areas, left/right) with Cz (feet/midline) are the core; adding FC1/FC2 or CP1/CP2 sharpens the spatial filter. A viable 8-ch MI montage: FC1, FC2, C3, Cz, C4, CP1, CP2, Pz.

train on reference data -> test on people?

or train on multiple people data and test on same + other people?

Feasibility at 8 channels. Good, especially for 2-class. The reference BCI Competition IV-2b dataset uses only **three** electrodes (C3, Cz, C4); a low-cost 8-channel OpenBCI Cyton study reached up to ~90% subject-specific online accuracy with EEGNet, and channel-selection work retains ~87% on the 22-channel IV-2a benchmark after dropping ~27% of sensors. 4-class at 8 dry channels is achievable but noticeably weaker.

Features / models. CSP or Filter-Bank CSP + LDA is the fast, strong baseline; Riemannian (covariance + MDM/tangent-space) is the robust state of the art and needs little tuning; EEGNet is the compact deep option and benefits from cross-subject pre-training.

Data & time. Subject-specific: 2-4 classes, ~20-40 trials/class of ~4 s with rest between, over 2-3 runs \~ 20-40 min/session; a second session improves robustness. Cross-subject or deep models want many subjects — pre-train on PhysioNet MMI (109 subjects, 64 ch) or IV-2a (9 subjects), then fine-tune. Classical models train in **seconds** on CPU; EEGNet in minutes (CPU) to seconds (GPU). Validate **trial/session-wise**, never window-wise, to avoid inflated numbers. [1,3,4,5,6,7,8,16,18,19]

4. ERD / ERS

What it is. Event-related desynchronization/synchronization is the neurophysiological mechanism MI exploits: a movement or its imagery causes a power *decrease* (ERD) in mu/beta before/during the act and a rebound *increase* (ERS, notably beta) afterwards, over sensorimotor cortex (Pfurtscheller). Treated as a paradigm, it is the simplest, most reliable sensorimotor read: detect *movement/imagery vs rest*, or left vs right, from band-power changes rather than a full multi-class decoder.

Best channels. C3, C4, Cz (identical to MI). **Feasibility:** High for 2-class detection at 8 dry channels — band-power on C3/C4 with a per-subject threshold is enough for a binary trigger. **Data:** small (~15-20 cued trials/class); computing the ERD curve needs only a rest baseline and cued epochs. **Control fit:** excellent as a low-rate *go/enable* or intent gate rather than a rich command channel. [1,2]

5. Emotion classification

What it decodes. Affective state, usually as valence (pleasant-unpleasant) and arousal (calm-excited), sometimes discrete emotions. Informative features are frontal alpha/beta asymmetry (F3/F4, Fp1/Fp2) and band-power / differential-entropy across theta-gamma, with temporal (T7/T8) and parietal sites contributing.

Best channels. Frontal first: Fp1, Fp2, F3, F4; then F7/F8 or T7/T8 and P3/P4. An 8-ch affective montage: Fp1, Fp2, F3, F4, F7, F8, P3, P4 (trades away sensorimotor/occipital coverage).

Feasibility at 8 channels. Low-Medium. Reduced-channel studies show 4-9 well-placed channels can approach full-montage performance, but reported accuracies are dataset- and subject-dependent (e.g. ~70% on DEAP valence/arousal quadrants; higher on SEED's cleaner 3-class task with 62 gel channels). Dry frontal electrodes are also the most blink/EOG-contaminated, and emotion labels are intrinsically noisy.

Data & time. This is the most data-hungry paradigm. It needs a validated *elicitation* protocol (film/music/IAPS) and self-report labels: DEAP = 32 subjects × 40 one-minute videos (32 ch@512 Hz); SEED = 15 subjects × 3 sessions of 15 clips (62 ch). Own subject-dependent data realistically means 30-60 min of labelled elicitation per person; cross-subject generalisation needs many subjects and still transfers poorly. Features (DE/PSD) + SVM/XGBoost train in seconds; deep models in minutes. **Verdict:** scientifically interesting but the weakest fit for reliable real-time control — defer. [9,10,11,12,20]

6. Other relevant signals

SSVEP (steady-state visual evoked potential). Gazing at a target flickering at a fixed frequency makes occipital cortex resonate at that frequency and its harmonics. **Channels:** O1, O2, Oz (+POz/Pz). **Feasibility:** High — the strongest, highest-ITR EEG signal and a top match for 8 channels; Canonical Correlation Analysis (CCA/FBCCA) is essentially calibration-free (~84% for 6 targets at 1 s windows in consumer-grade setups). **Cost:** needs an external flickering stimulus (screen/LEDs). Ideal hands-free *menu/target selection*. [13,14,21]

P300. A rare, attended stimulus in an oddball sequence evokes a ~300 ms positive parietal deflection. **Channels:** Pz, Cz, Fz (+P3/P4). **Feasibility:** Medium at 8 dry channels — single-trial P300 is small, so several

stimulus repetitions are averaged, which slows it. Good for discrete speller/menu selection; short calibration. [15]

Occipital alpha (eyes-closed). Alpha (8-12 Hz) at O1/O2 rises sharply on eye closure. **Feasibility:** High and training-free (threshold on the alpha ratio — already implemented in biopipe's monitor). Useful as an idle/attention switch or a coarse binary command.

Error-related potentials (ErrP). A frontocentral (FCz/Cz) response when the user perceives a wrong action. **Feasibility:** Medium; needs moderate calibration. High *control* value as a corrective/veto layer over any decoder — the system can cancel a command the brain flags as wrong. [17]

Auxiliary (not cortical): EOG blinks / jaw. Large, trivially detected on frontal channels; usable as reliable discrete triggers, but they are ocular/muscular, not brain signals — use only as a fallback switch.

7. Training requirements at a glance

Paradigm	Public datasets (subjects, channels)	Own-data guidance	Model + train time
MI (2-4 class)	BCI IV-2a (9s, 22ch), IV-2b (9s, 3ch: C3/Cz/C4), PhysioNet MMI (109s, 64ch)	1+ subj; 20-40 trials/class ×4s; 20-40 min/session	CSP+LDA / Riemannian / EEGNet; sec-min
ERD/ERS (2-class)	IV-2b; PhysioNet MMI	1+ subj; ~15-20 cued trials/class + rest	Band-power + LDA/threshold; sec
Emotion (VA)	DEAP (32s, 40×1min, 32ch), SEED (15s, 62ch)	many subj OR 30-60 min elicitation/subj	DE/PSD + SVM/XGBoost / deep; sec-min
SSVEP	Tsinghua benchmark; own flicker set	calibration-free; secs/target to verify	CCA / FBCCA; none
P300	own oddball recordings	1 subj; ~5-15 reps/selection, short cal.	xDAWN+LDA / Riemannian; sec

Notes. Classical pipelines (CSP/Riemannian/CCA/xDAWN + LDA/SVM) train in seconds on a laptop CPU and are the right default here; deep nets (EEGNet) only pay off with cross-subject pre-training. Always split train/test by **trial or session**, report per-class scores, and expect a 2-4 min impedance/comfort setup per session with dry electrodes.

8. Recommended 8-channel montage

No single 8-electrode layout is optimal for every paradigm, so choose per intended use. Two practical options:

• **Sensorimotor-first (MI / ERD / ErrP + SSVEP):** FCz, C3, Cz, C4, CP1, CP2, O1, O2 — covers the motor strip for MI/ERD, FCz for ErrP, and O1/O2 for SSVEP and alpha. **Recommended** for a control-oriented hybrid.

• **Broad/mixed (adds affective + P300):** F3, F4, C3, Cz, C4, Pz, O1, O2 — frontal for emotion/ErrP, central for MI/ERD, Pz for P300, occipital for SSVEP/alpha. More general, weaker per-paradigm.

biopipe's 10-20 selector lets a session store exactly which electrode fed which channel, so different montages coexist and only compatible recordings are pooled for training.

9. Path to task control — recommendation

Ranked by reliability-per-effort on this 8-channel dry rig, and by how cleanly each yields a discrete command:

1. **SSVEP** — fastest, most robust, near-zero training. Best first EEG command channel (hands-free menu / mode / target select). Needs a flicker stimulus.
2. **ERD/ERS + MI** — no external stimulus; a 2-class C3/C4 detector gives a reliable left/right or go/rest command, extensible toward 4-class MI later.
3. **Occipital alpha + ErrP** — lightweight support signals: alpha as an idle/enable switch, ErrP as a corrective veto that raises the safety of any decoder.
4. **Emotion** — defer: high data cost and low real-time reliability make it unsuitable as a primary control channel now; revisit as an offline/adaptive signal.

Fit with biopipe. The eventual controller follows the hybrid, decision-level-fusion scheme in the project roadmap: **EMG** carries fast primary commands (already working), **EEG-SSVEP or MI** provides a hands-free selection/mode channel, occipital **alpha** gates idle/active, and **ErrP** can veto wrong actions. Each modality does what it is best at and a weak read on one channel is out-voted rather than sinking the command.

Proposed build order for the EEG training UI/backend: (1) generalise the existing recording + band-power pipeline to a cued, windowed *2-class ERD/MI* trainer (reuses biopipe's multi-label model and schema); (2) add a *SSVEP* mode (flicker config + CCA, calibration-free); (3) add supporting *alpha*/*ErrP* detectors; (4) treat *emotion* as a later, offline research track.

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